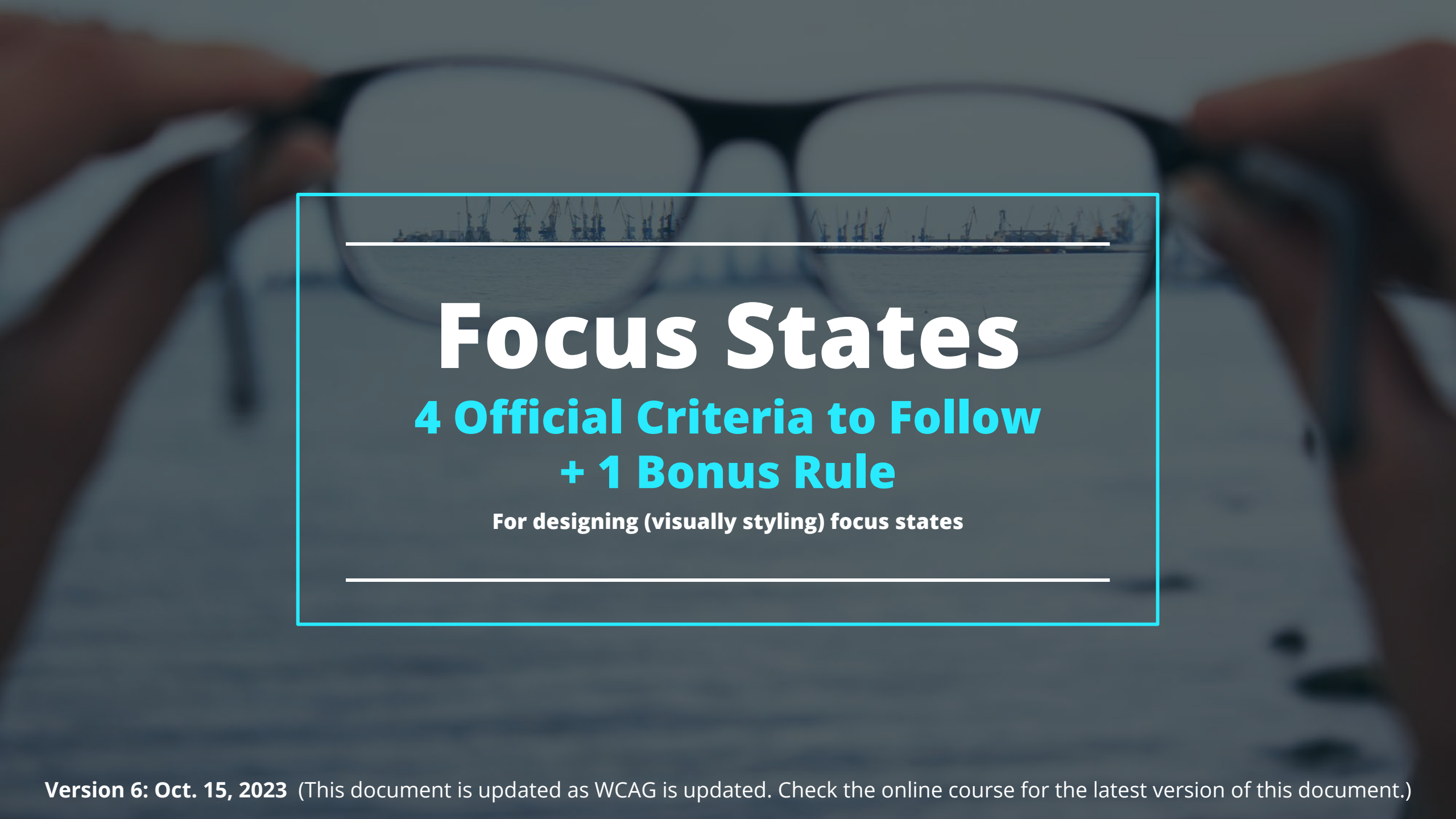




Focus States

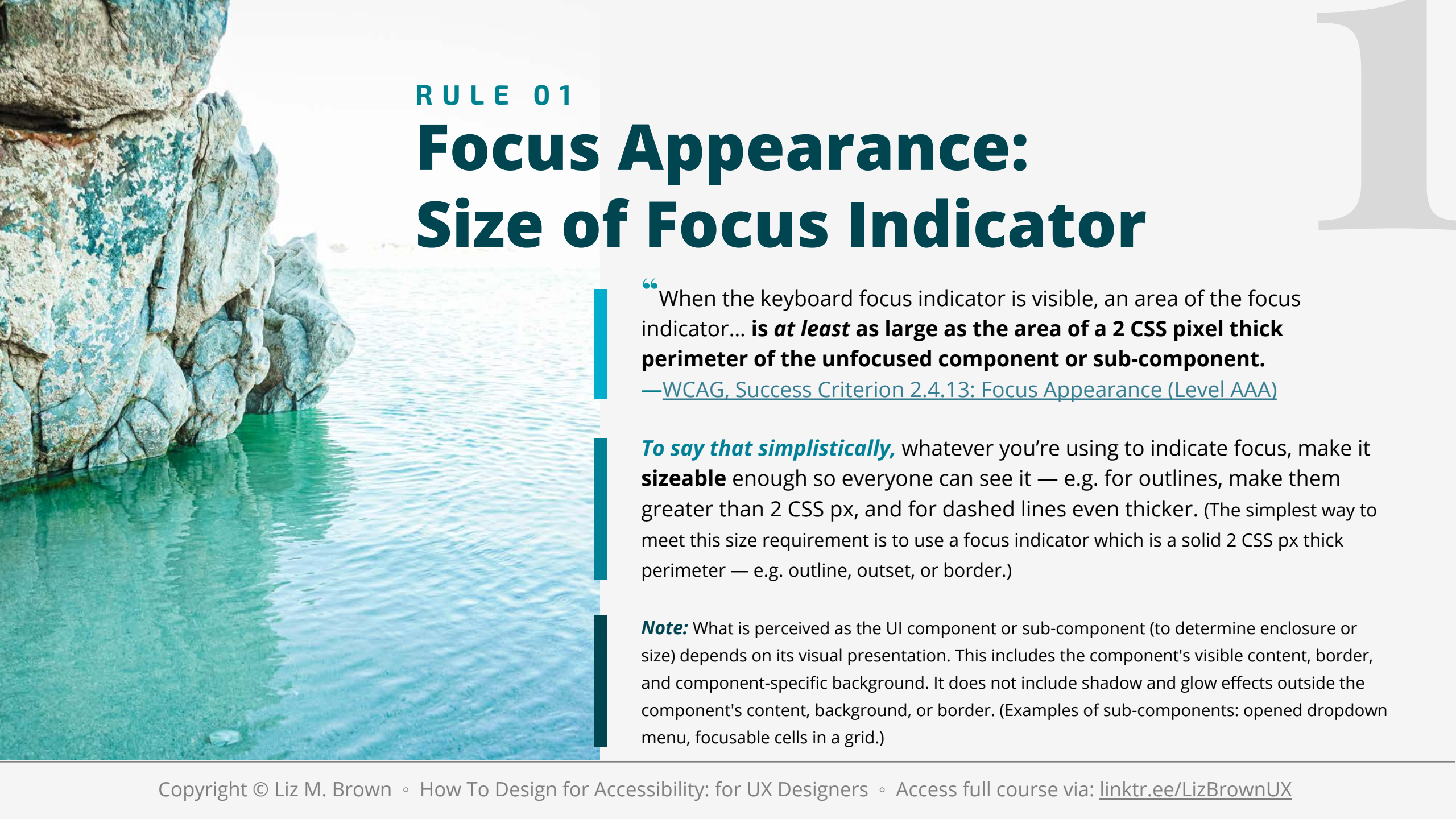
4 Official Criteria to Follow + 1 Bonus Rule

For designing (visually styling) focus states

WCAG says that when UI components receive keyboard focus, the subsequent

4 criteria need ↩ “Rules” #1 - #4 to be true for accessibility compliance





RULE 01

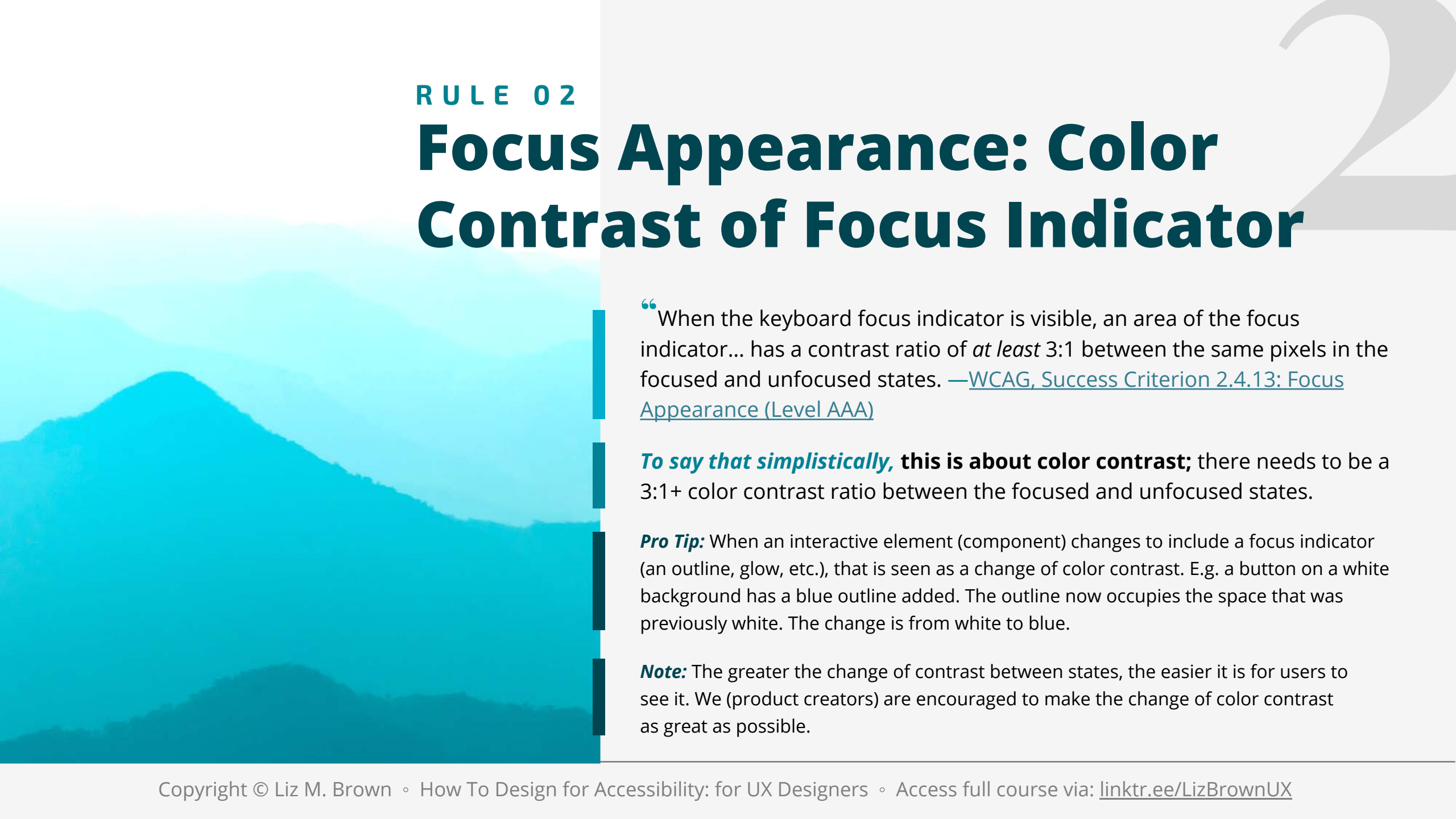
Focus Appearance: Size of Focus Indicator

“When the keyboard focus indicator is visible, an area of the focus indicator... **is *at least* as large as the area of a 2 CSS pixel thick perimeter of the unfocused component or sub-component.**

—[WCAG, Success Criterion 2.4.13: Focus Appearance \(Level AAA\)](#)

To say that simplistically, whatever you're using to indicate focus, make it **sizeable** enough so everyone can see it — e.g. for outlines, make them greater than 2 CSS px, and for dashed lines even thicker. (The simplest way to meet this size requirement is to use a focus indicator which is a solid 2 CSS px thick perimeter — e.g. outline, outset, or border.)

Note: What is perceived as the UI component or sub-component (to determine enclosure or size) depends on its visual presentation. This includes the component's visible content, border, and component-specific background. It does not include shadow and glow effects outside the component's content, background, or border. (Examples of sub-components: opened dropdown menu, focusable cells in a grid.)



RULE 02

Focus Appearance: Color Contrast of Focus Indicator

“When the keyboard focus indicator is visible, an area of the focus indicator... has a contrast ratio of *at least* 3:1 between the same pixels in the focused and unfocused states. —[WCAG, Success Criterion 2.4.13: Focus Appearance \(Level AAA\)](#)”

To say that *simplistically*, this is about color contrast; there needs to be a 3:1+ color contrast ratio between the focused and unfocused states.

Pro Tip: When an interactive element (component) changes to include a focus indicator (an outline, glow, etc.), that is seen as a change of color contrast. E.g. a button on a white background has a blue outline added. The outline now occupies the space that was previously white. The change is from white to blue.

Note: The greater the change of contrast between states, the easier it is for users to see it. We (product creators) are encouraged to make the change of color contrast as great as possible.



RULE 03

Non-Text Contrast: Contrast Against Adjacent Colors

“Focus indicators are visual information required to identify a state of a user interface component. That means that they are also subject to [1.4.11 Non-text Contrast \(Level AA\)](#). (Which states, **“The visual presentation of UI components and graphical objects have a contrast ratio of at least 3:1 against adjacent color(s).”**) This requires that the visual focus indicator for a component must have sufficient contrast against the adjacent colors when the component is focused. —[WCAG, Success Criterion 2.4.13: Focus Appearance \(Level AAA\)](#)

To say that simplistically, **this is about color independence;** there needs to be a 3:1+ color contrast ratio between the focus indicator and the adjacent colors. E.g. if the focus indicator is a blue outline, and it's on a white page background, then the color contrast would be measured between the blue and white.

Note: If the focus indicator sufficiently contrasts with the background, it would not need to also contrast with the component if the outline was very thick; probably greater than 3 or 4px (making the overall size of the component appear noticeably larger). But even still, this is not the ideal design (examples forthcoming).



RULE 04

Focus Not Obscured: Focused Component Not Hidden

“*[Minimum requirement:]* When a user interface component receives keyboard focus, the component is **not entirely hidden** due to author-created content. —[WCAG, Success Criterion 2.4.11: Focus Not Obscured \(Minimum\)](#) (Level AA)

[Enhanced requirement:] When a user interface component receives keyboard focus, **no part of the component is hidden** by author-created content. —[WCAG, Success Criterion 2.4.12: Focus Not Obscured \(Enhanced\)](#) (Level AAA)

To say that simplistically, a “visible focus state” isn’t so visible if it gets covered up by other stuff on the page.

Note: Typical types of content that can overlap focused items are sticky footers, sticky headers, and non-modal dialogs. As a user tabs through the page, these layers of content can hide the item receiving focus, along with its focus indicator.

BONUS
RULE 05

Befitting Signifiers Used To Indicate Focus

The presentation (styling, micro animation, etc.) used to indicate the focus state, should use befitting signifiers to accurately communicate its meaning to users. (Signifiers are: Signals that communicate an element's intended use. They can be visual, auditory, tactile, etc.)

To say that *simplistically*, if an interactive element is supposed to be visually indicating that it's focused, does it in fact look focused? Or does it look like another state or another element? Usability testing can answer this. (Usability is good for people *with* and *without* disabilities.)

Note: As of yet, there is no specific WCAG page to cite for this one. However, Don Norman, co-founder of the Nielsen Norman Group (a source of authority on usability) provides a reputable and reliable source we can reference [about Signifiers: jnd.org/signifiers not affordances](https://www.nngroup.com/articles/signifiers-not-affordances/).

Additional WCAG Criteria

**Not explicitly related to visual styling,
but important to know and implement:**

Focus Visible

“Any keyboard operable user interface has a mode of operation where the keyboard focus indicator is visible. —[WCAG, Success Criterion 2.4.7: Focus Visible \(Level AA\)](#)

Focus Order

“If a Web page can be navigated sequentially and the navigation sequences affect meaning or operation, focusable components receive focus in an order that preserves meaning and operability. —[WCAG, Success Criterion 2.4.3: Focus Order \(Level A\)](#)

On Focus

“When any user interface component receives focus, it does not initiate a change of context. —[WCAG, Success Criterion 3.2.1: On Focus \(Level A\)](#)

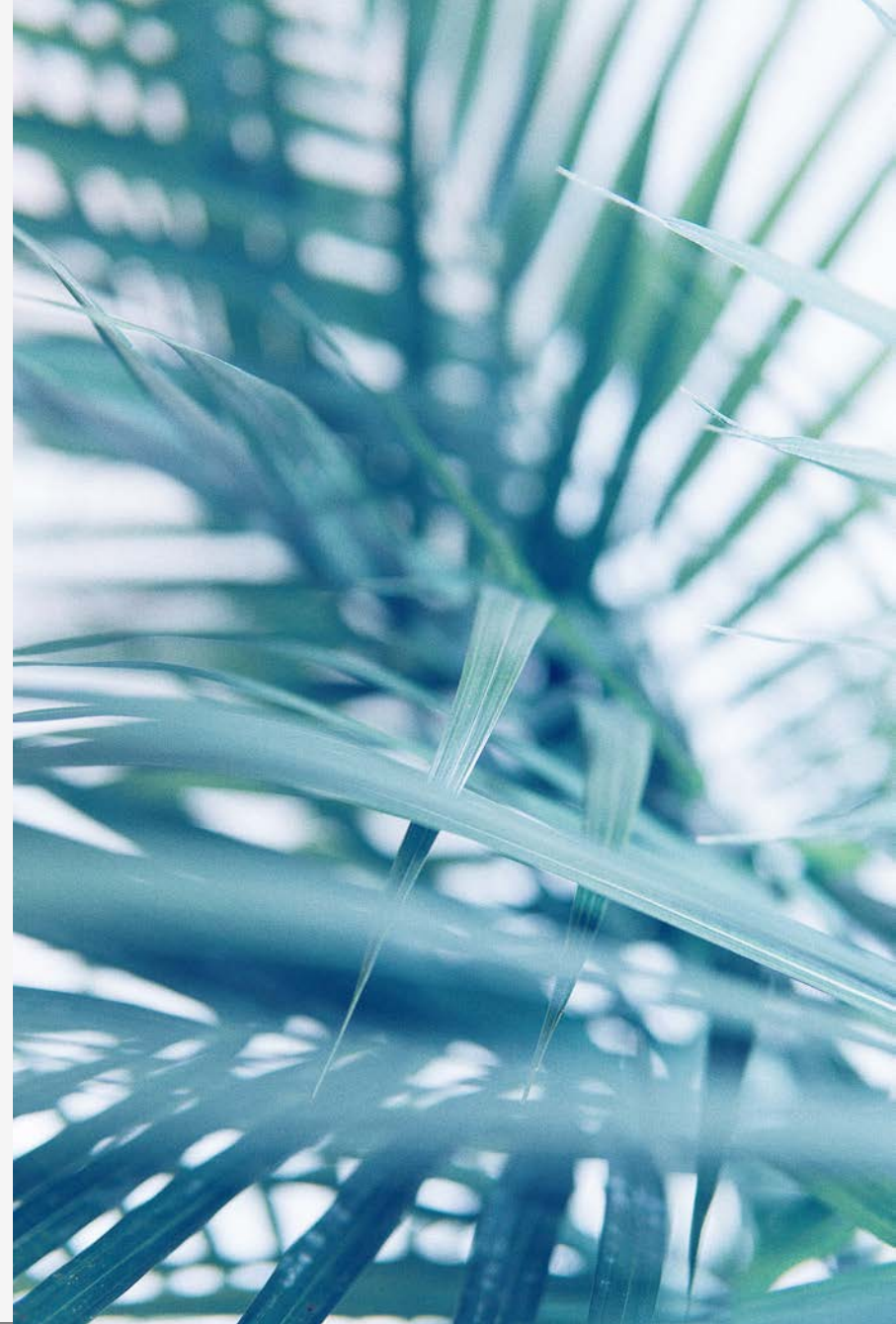
Content on Hover/Focus

“Where receiving and then removing pointer hover or keyboard focus triggers additional content to become visible and then hidden, the following are true: (1) Dismissible — A mechanism is available to dismiss the additional content without moving pointer hover or keyboard focus, unless the additional content communicates an input error or does not obscure or replace other content; (2) Hoverable — If pointer hover can trigger the additional content, then the pointer can be moved over the additional content without the additional content disappearing; (3) Persistent — The additional content remains visible until the hover or focus trigger is removed, the user dismisses it, or its information is no longer valid. —[WCAG, Success Criterion 1.4.13: Content on Hover or Focus \(Level AA\)](#)

10 pass & fail Examples

NOTE:

I have not included examples of rule #4 (Focus Not Obscured) since achieving it is fairly self-explanatory.



EXAMPLE 1: PASS

Solid black outline

✅ Size of Focus Indicator

A solid outline, greater than 2 CSS px, around the whole component would pass the size requirement. These examples have a 3px outline (approximately). This focus styling (on these examples) passes.

✅ Color Contrast (Focused vs. Unfocused)

For the contrast requirement, the area around these interactive elements is white, so a change from that white background to the black outline that now occupies that space, must meet or exceed 3:1. At 21:1, this passes.

✅ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The black outline should contrast with the colors it's touching at 3:1+. The bright blue (of the primary buttons and selected radio buttons) is 4.27:1 against the black outline. (Also, it exceeds 2px.) This passes. **Caution:** Because the black outline would touch everything it encircles, you'd have to check the colors in every instance; very tedious; leave a gap instead (an example of this is coming up).

✅ Befitting Signifiers

A thick outline (of some sort) has classically been used to signify the focus state. Therefore it would not be unexpected. This passes.

Default

Focused

Default

Focused

I'm a Focused link, aloha!
I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☒ Focused

☐ Default

☒ Focused

Notice: The focus outline is around the entire clickable area. We make both the radio circle **and** label clickable, because we're not jerks 😊

EXAMPLE 2: FAIL

Blue outline & glow

✅ Size of Focus Indicator

The blue outline is solid, it goes around the entire element, and at 2px (not counting the glow) it meets the 2px minimum requirement. This passes the size requirement.

❌ Color Contrast (Focused vs. Unfocused)

The change is from a white background to a light-blue outline (we ignore the glow). The contrast between those two colors is 1.92:1. Because it does not exceed 3:1, this fails.

❌ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The light-blue focus outline against white fails to meet/exceed 3:1 (it's only 1.92:1). Where the light-blue touches a cobalt-blue element, it also fails to meet/exceed 3:1 (it's only 2.54:1).

✅ Befitting Signifiers

This is a big ol' winner in the signifier category. How many focus states have you seen styled with a glowing blue aura like this, right! This styling is heavily associated with focus states (thanks in no small part to browsers' default focus states). Too bad it's not accessible.



I'm a Focused link, aloha!
I'm a Default link, hola!

Default Field:

Focused Field:

- ☐ Default
- ☐ Focused
- ☐ Default
- ☐ Focused

Do you notice
what's crummy
here?

EXAMPLE 3: PASS

Black outline, blue glow

✅ Size of Focus Indicator

The focus indicator is a 2px solid black outline that also has a blue glow. This passes the size requirement. (When the black outline is only 1px, it's difficult to see against the blue buttons and input field borders. And as I explain in a moment, we are disregarding the blue glow.)

✅ Color Contrast (Focused vs. Unfocused)

The white space around the elements changes to a black outline & blue glow. White to black exceeds 3:1 and passes. White to blue does not meet 3:1 and fails. We disregard the blue glow (consider it decoration). But the black outline alone is sufficient, so it enables this to pass.

✅ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The black outline (at only 2px) should contrast with the colors it's touching at 3:1+. Since the blue glow failed color contrast, we are still disregarding it (turn it off in your design to help you judge this). Black against the bright-blue buttons is 4.27:1, pass. Black against the very-dark-gray border of the input field, fail. To compensate, the focus indicator (black outline) needs to be at least 2px thick to distinguish it. For this reason, the black outline is 2px throughout, not 1px.

✅ Befitting Signifiers

Do we include the glow in our assessment of Befitting Signifiers? No, we disregard it here as well. We imagine ourselves as having low vision or color blindness — We don't pick and choose when to not see the blue glow. So we assess Befitting Signifiers based on the 2px black outline alone. And as explained in a previous example, it passes.

Default

Focused

Default

Focused

I'm a Focused link, aloha!
I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☒ Focused

☐ Default

☒ Focused

EXAMPLE 4: PASS

Outline with a gap

✓ Size of Focus Indicator

Two examples here: The first set of buttons has a cyan outline for the focus indicator, the second set has a cobalt outline. For both, the 2px solid blue outline (with a soft glow) meets the minimum 2 CSS px thickness requirement (around the entire component) and therefore passes the size requirement. **Note:** the gap should be 2 CSS px or larger to be noticeable and count as a “gap”.

✓ Color Contrast (Focused vs. Unfocused)

In the first set of buttons, the change is from white to cyan — at 3.14:1 this passes (it meets/exceeds 3:1). (Again, we disregard the decorative glow.) In the second set of buttons, the change is from white to cobalt — at 6.58:1 it also passes.

✓ Adjacent Color Contrast (Focus Indicator Against Background/Component)

Thanks to the noticeable (greater than 2px) gap between the outline and the element it surrounds, and the background color showing through the gap, we now only have to measure adjacent contrast against one color — white (for all interactive elements). Both cyan (top pair of examples) and cobalt (bottom pair of examples) against white exceed 3:1. Both examples pass.

✓ Befitting Signifiers

An outline is a classic signifier for the focus state. This passes.



EXAMPLE 5: PASS

2-color double outline

✓ Size of Focus Indicator

The double-outline has a white inner stroke and a blue outer stroke, each 2px thick (exceeding the required 2px for solid lines). It also has a gap between the outline set and the element that exceeds the necessary 2px to be noticeable. Overall, this passes the size req.

✓ Color Contrast (Focused vs. Unfocused)

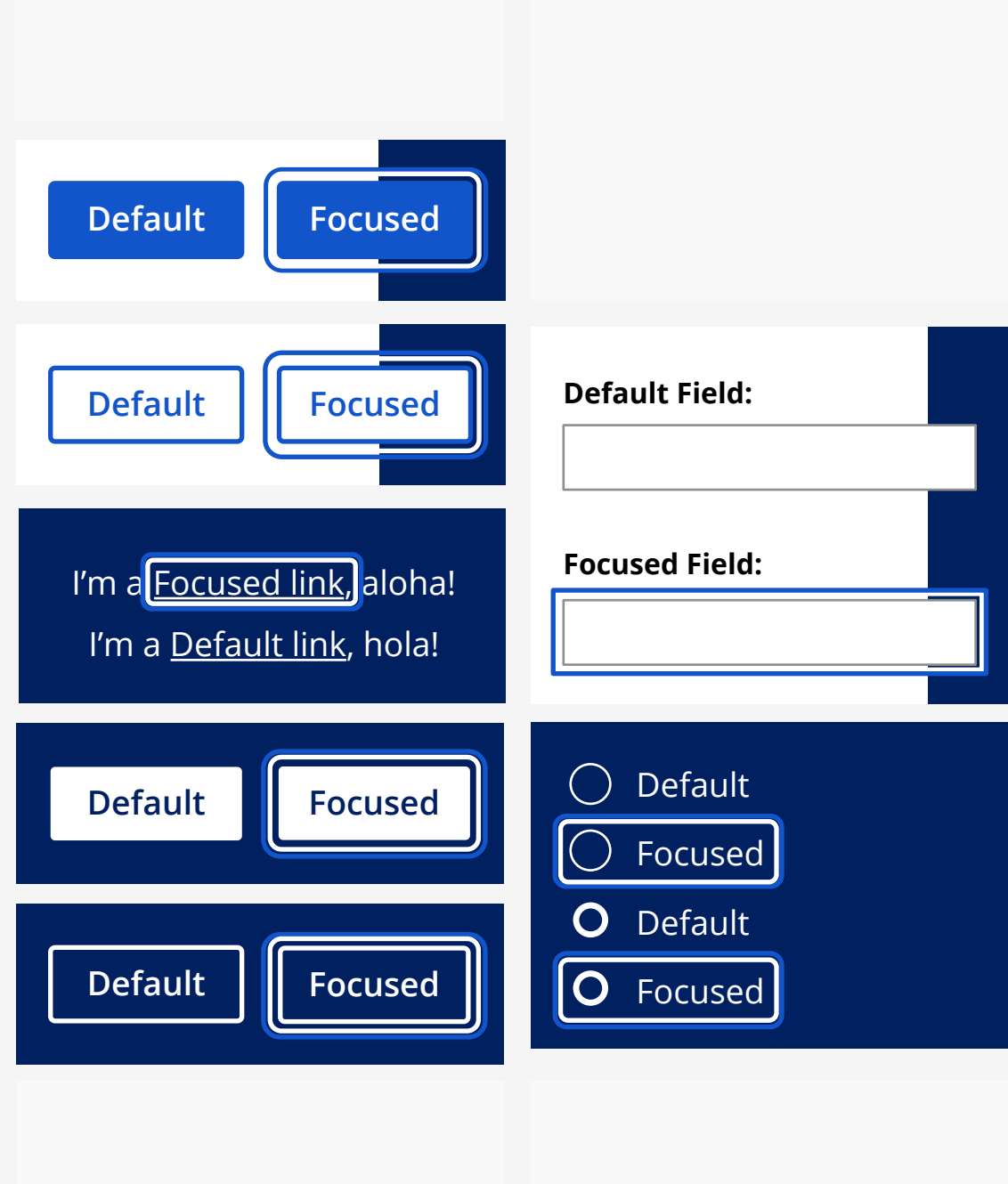
The change is from a white (or dark-blue) background to a white & blue double outline. When the white outline is visually consumed by the background, the blue outline carries the weight — white to blue exceeds 3:1 and passes. When the blue outline is visually consumed by the background, the white outline carries the weight — dark-blue to white exceeds 3:1 and passes. **A similar styling strategy:** a 4px dashed line, but instead of alternating between dashes and gaps, it's alternating between blue and white dashes (no gaps).

✓ Adjacent Color Contrast (Focus Indicator Against Background/Component)

White against blue or blue against white, they exceed 3:1 and pass. Since we've used a gap between the outline set and the element it surrounds, we've saved this styling from failing. For example, without the gap the white buttons (on the dark background) would have visually consumed the white outline, and the dark background would have visually consumed the blue outline, leaving no noticeable focus outline, thereby failing. And same problem if the inner outline was blue instead of white, it would just fail on the blue buttons instead.

✓ Befitting Signifiers

Outlines are expected signifiers of focus. This passes.



EXAMPLE 6: FAIL

Thin dashed outline

✗ Size of Focus Indicator

A 2 CSS px dashed outline around the whole component fails the size requirement, as it is roughly 50% of the surface area of a solid line. (In most browsers a dashed line will have roughly half the number of pixels due to gaps.)

✓ Color Contrast (Focused vs. Unfocused)

A change from the white background to the black dashed outline that now occupies that space, exceeds the required 3:1. This passes.

✗ Adjacent Color Contrast (Focus Indicator Against Background/Component)

In every instance, the black dashed outline needs to contrast with the colors it's touching at 3:1+. In these examples, it passes against some and fails against others (e.g. the selected radio buttons, the secondary buttons.) It's tedious to check this on an entire website's worth of interactive elements, so a gap is recommended.

✓ Befitting Signifiers

A dashed outline is another common signifier of focus. Some dashed lines may be hard to see. But dashed lines in general pass Befitting Signifiers.

Default

Focused

Default

Focused

I'm a Focused link, aloha!
I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☒ Focused

☐ Default

☒ Focused

EXAMPLE 7: PASS

Thick dashed outline

✅ Size of Focus Indicator

The page background is dark blue with white text and interactive elements. When focus lands on an element, it's outlined with a bright yellow, 4px, dashed line. This passes the size requirement for dashed lines.

✅ Color Contrast (Focused vs. Unfocused)

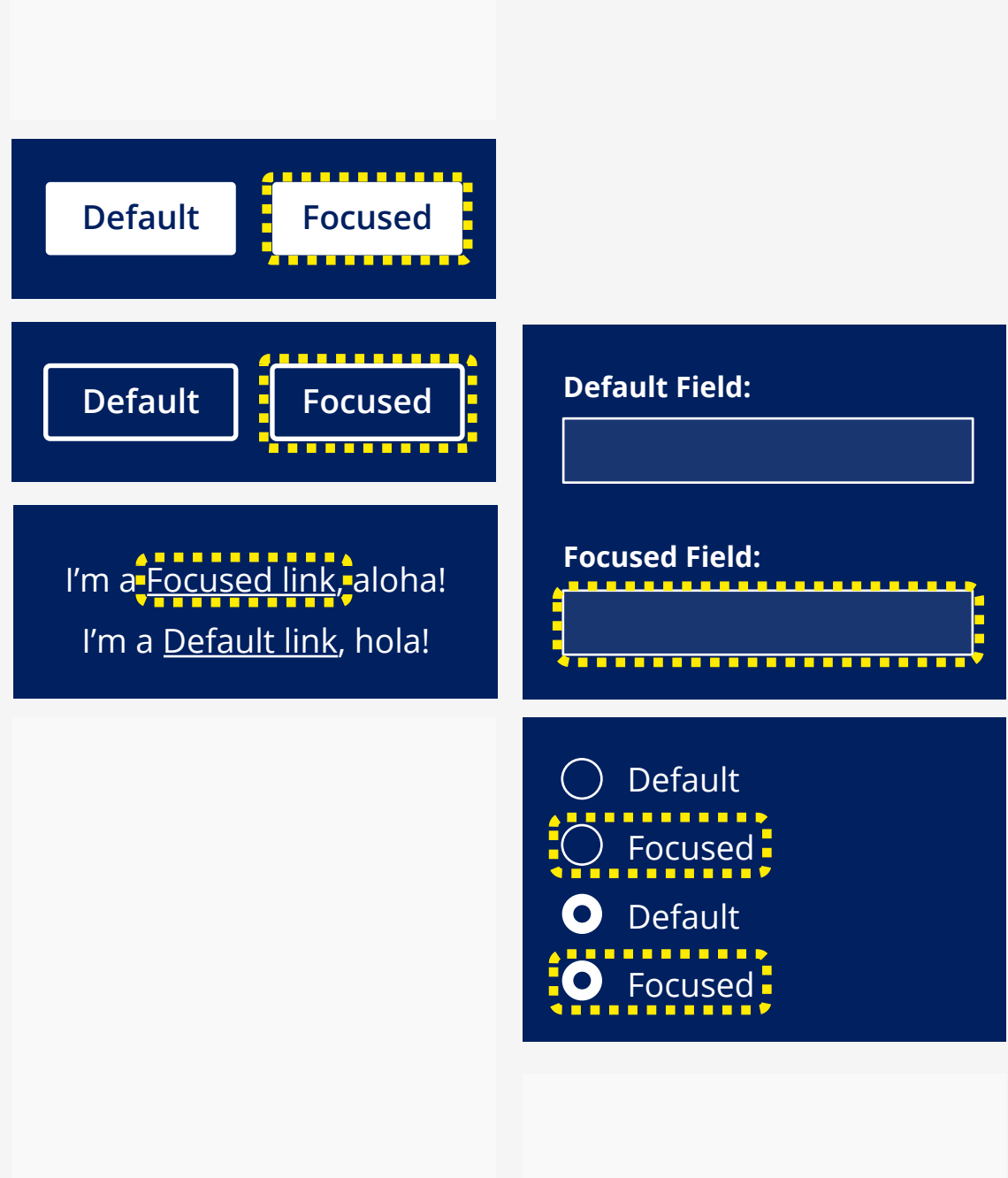
The area around these interactive elements is dark blue, so the change from that background color (dark blue) to the focus indicator color (bright yellow) that now occupies that space, must have a ratio of 3:1+. At 12.37:1, this passes.

✅ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The bright yellow against white interactive elements fails (1.23:1), but thanks to the dashed line's thickness and gaps, there's enough of the dark blue background showing through to make the outline distinguishable and not blend in with the white elements.

✅ Befitting Signifiers

A thick outline (of some sort) has classically been used to signify the focus state, and would not be unexpected. This passes.



EXAMPLE 8: FAIL

Highlight color (aka change in background color)

✅ Size of Focus Indicator

Focus is indicated by a yellow highlight that fills the background of the focused element. Since it highlights the entire clickable area, this passes the size requirement. **Side note:** For a focused input field, a blinking cursor (vertical line inside the field) would fail the size requirement, and can not be relied on alone to indicate focus. (Note: `textarea` elements, and the like, are considered user interface components and need to have focus states. —[WCAG 2.2 Focus Appearance](#).)

❌ Color Contrast (Focused vs. Unfocused)

The change is from a white background to pale yellow. At 1.04:1, this fails to meet the required 3:1 contrast ratio. **Reminder:** the text color must maintain a 4.5:1+ ratio with the new background color — in these examples it does.

❌ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The yellow highlight needs to contrast with the colors it's touching at 3:1+. Yellow against white — **fail**. Yellow against cobalt — pass. Yellow against black — pass. Yellow against dark-gray — pass.

❌ Befitting Signifiers

When a user clicks and drags their cursor over something, that's typically when a highlight is seen. If we stole that styling to use it for our focus indicators, it may lead to confusion. Also, this is not very flexible styling; how do you "highlight" a dark-colored element? Too drastic of a change to make the highlight work causes the element to look like something it's not (like the focused primary button in this example, which now looks like a secondary button).

Default

Focused

Default

Focused

I'm a Focused link, aloha!

I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☒ Focused

☐ Default

☒ Focused

EXAMPLE 9: FAIL

Solid vs. hollow fill (aka change in background color)

✅ Size of Focus Indicator

Upon receiving focus, the entire background color changes — this passes the size req.

✅ Color Contrast (Focused vs. Unfocused)

The default background color and the new one need a 3:1+ contrast ratio with each other for the color change to be noticeable. In all the examples, the shift from blue/white to white/blue achieves a 4.9:1 ratio, exceeding the requirement. Also, the text must maintain a 4.5:1+ ratio with its background. To achieve this, when the background color changes the text color does as well. This passes.

✅ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The new background color must contrast 3:1+ with its adjacent colors. In these examples it's white against blue and blue against white — they pass. (What about the blue against the dark-gray border of the input field and unselected radio button? The dark-gray is visually absorbed and becomes part of the blue shape, but no information / identifiability is lost, so it still passes. See [WCAG Non-Text Contrast](#) for more details on this.)

❌ Befitting Signifiers

Using this styling causes some elements to change signifiers. For example, the focused primary button now looks like a secondary button, the focused input field now looks like a solid rectangle and not something you can type into, and so on. This styling is not clearly signifying “focus”. It fails.

Default

Focused

Default

Focused

I'm a Focused link, aloha!

I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☒ Focused

☐ Default

☐ Focused

EXAMPLE 10: FAIL

Thick underline

✗ Size of Focus Indicator

A thick (3px) underline, that's almost as long as the longest side of the button, does not meet the size requirement. The total amount of pixels in the underline adds up to fewer pixels than the the total amount of pixels in a 2px continuous perimeter.

[WCAG 2.2](#) says the focus indicator needs to be "at least as large as the area of a 2 CSS pixel thick perimeter," and it defines "perimeter" as a "continuous line forming the boundary of a shape not including shared pixels, or the minimum bounding box, whichever is shortest." It also provides the formula to do the math, if you're interested.

✓ Color Contrast (Focused vs. Unfocused)

For the primary buttons, the dark-blue area that's now occupied by the yellow underline is a change from dark-blue to yellow, at 12.37:1 this passes. For the other elements, the white area that's now occupied by the dark-blue underline is white to blue which also exceeds the 3:1 requirement. This passes.

✓ Adjacent Color Contrast (Focus Indicator Against Background/Component)

The underline must contrast with the colors it's touching at 3:1+. The yellow underline against the dark blue passes. The dark blue against the white also passes.

? ✓ Befitting Signifiers

This styling is not common for signifying focus. From the buttons we can imagine how this styling would look on main navigation items or tabs; it may potentially be mistaken for the "current" state. But judging only the examples shown here, it's unique, but it could pass. **However**, usability testing would be needed to know for sure if this passes or fails Befitting Signifiers.

Default

Focused

Default

Focused

I'm a Focused link, aloha!

I'm a Default link, hola!

Default Field:

Focused Field:

☐ Default

☐ Focused

☒ Default

☒ Focused

Sources

- The 4 official criteria (what I've called "rules") and the 4 additional criteria are from the Web Content Accessibility Guidelines (WCAG) 2.2
- All photography is from: Unsplash.com

Notices

Copyright © Liz M. Brown. All rights reserved.

- This document is copyright protected and for your personal use only.
- As a condition of use, you agree to **retain all copyright and notices**, and to not modify any of the material in any manner.

No reposting, reusing, selling, or editing.

- Do not distribute, post, transmit, amend, sell, re-purpose, reproduce, duplicate, or re-use any part of this document without direct written consent from the author.
- Any violations may result in civil and/or criminal liabilities.

Limitation of liability.

- This document is for educational purposes only and does not constitute legal, financial, medical, or professional advice.
- Under no circumstances will any blame, legal responsibility, or liability be held against the author for any damages, reparation, or monetary loss due to the information contained within this document and/or its use, directly or indirectly.

Use at your own risk.

- All effort has been made to present accurate, up-to-date, reliable, complete information. The author takes no responsibility or liability for any errors or omissions. The information is provided on an "as is" basis with no guarantees or warranties of any kind whatsoever, expressed or implied.
- The author is not responsible for your use of the information contained, or linked to, in this document, which has been derived from various sources. Please consult a licensed professional before attempting any techniques outlined.